Karkinos Healthcare: Making cancer detection and care delivery highly available with a scalable cloud infrastructure

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About Karkinos Healthcare

Karkinos Healthcare is a technology-led oncology platform that is focused on designing and delivering bespoke solutions for cancer care and creating 'cancer centers without walls' to reduce the burden of cancer for all Indians. It is supported by a founding team with deep technology, healthcare and oncology experience, and a panel of eminent clinical advisors in India and abroad.

Industries: Healthcare

With Google Clou Karkinos Healthc helped hundreds thousands of unc Indians across th to enjoy easier ac

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Location: India

(https://workspace.google.com/products/docs/)

Google Cloud results

- Support a platform that has delivered early cancer detection to over 400,000 Indians
- Enable over 5,000 cancer risk assessment surveys per day
- Process 10,000 genomics sequences per year that involve over 1.5 terabytes of

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About Persistent Systems

Persistent is a trusted Digital Engineering and Enterprise Modernization partner, combining deep technical expertise and industry experience to help our clients anticipate what's next and answer questions before they're asked. Our offerings and proven solutions create unique competitive advantage for our clients by giving them the power to see beyond and rise above.

Google Cloud (https://cloud.google.com/)

Tell us your challenge. We're here to help.

Cloud AutoML (https://cloud.google.com/automl/)

Docs (https://workspace.google.com/products/docs/)

Cloud Healthcare API (https://cloud.google.com/healthcare-api/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud SQL (https://cloud.google.com/sql/)

Cloud Storage (https://cloud.google.com/storage/)

About Persistent Systems

data per day
- Onboard 70 partner clinics across India to bring oncology support closer to the patient
- Provide an upskilling opportunity for doctors to deliver oncological services

Sheets (https://workspace.google.com/products/sheets/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Google Workspace (https://workspace.google.com/)

Google Cloud Armor (https://cloud.google.com/armor/)

Compute Engine (https://cloud.google.com/compute/)

Slides (https://workspace.google.com/products/slides/)

Every year, more than a million Indians are diagnosed with cancer (https://www.thinkglobalhealth.org/article/india-needs-cancer-care-outside-its-big-cities). This staggering number, combined with limited access to healthcare, has become a real concern in the country. In an attempt to lower the rate of advanced-stage cancer, Karkinos Healthcare (https://www.karkinos.in/) (Karkinos), a technology-led oncology platform, is on a mission to dramatically increase early detection and diagnosis to reduce the burden of cancer on the nation's health infrastructure, and the wallets of every Indian.

"Expansion of physical infrastructure and available manpower has its limitations. Technology not only helps to extend the resources, but the most important healthcare requirement, which is ensuring standardization and quality of care. A digitally connected cancer care can create the much needed milieu for innovation of next generation interventions."

—Dr Moni Abraham Kuriakose, Co-founder and Medical Director, Karkinos

"With over four million cancer patients seeking care at any given time in India, which is increasing steadily, the care availability- need gap continues to widen," says Dr Moni Abraham Kuriakose, Co-founder and Medical Director at Karkinos. "Expansion of physical infrastructure and available manpower has its limitations. Technology not only helps to extend the resources, but the most important healthcare requirement, which is ensuring standardization and quality of care. A digitally connected cancer care can create the much needed milieu for innovation of next generation interventions."

Based on Karkinos' experience, many people only begin treatment some nine months after they are first diagnosed due to a lack of knowledge on what to do next. That's where Karkinos' end-to-end platform, which unites patients, partner clinics, and clinicians, can help.

Building an end-to-end platform for oncological service delivery

According to Raja Sekhar Kommu, the company's co-founder and Chief Technology Officer, Karkinos built its platform on Google Cloud (https://cloud.google.com/) from the beginning due to three major drivers in the company's architecture design philosophy.

First, the platform needed to conduct low-cost, population-scale risk assessments, which is huge for a country like India. These assessments are used to stratify cancer risk for a particular individual.

"By definition, whatever we build must scale, and it can't be a small, isolated app on a monolithic architecture. Furthermore, how you conduct a risk assessment in Kerala differs greatly from Chhattisgarh. These are two very different environments with local considerations that factor into the assessments, including up to a dozen languages," Kommu reveals.

These assessments comprise up to 70 programmatically generated questions derived from the expertise of Karkinos' clinicians and various country-level and healthcare industry guidelines. Since the company's founding in 2020, over 400,000 Indians have taken the surveys and have had their risks categorized.

"Of those assessed, 12% have been identified as having a high risk for cancer. They are then navigated through the Virtual Command Center on our platform and are nudged to undergo more comprehensive screening after the assessment," Kommu explains.

As internet availability is not uniform across India, Karkinos' platform needed to be designed from day one to accommodate inconsistent user experiences on the ground, especially in remote towns.

Second, Karkinos' platform needed to ingest and process large volumes of genomics data derived from the tissues collected from the patient's entire treatment journey. While fewer patients are involved here, the platform needs to process many samples per patient.

"With genomics data, for example, one run for genomics generates 3.2 terabytes of data in 48 hours. We are on the path to increasing our genomics sequencing load to 10,000 sequences per year. We currently conduct 4,000 risk assessments a week, but that volume is also set to increase," Kommu says.

This genomics data helps Karkinos create Indian population specific cancer atlases that are useful to improve case outcomes and survival. Early-stage cervical cancer, for example, only involves a 20-second treatment by thermal ablation to treat the cancer.

Moreover, the patient's out-of-pocket expense is very small, which is an important consideration in an underinsured country like India.

Third, Google Cloud did not require Karkinos to tie itself to proprietary software. Instead, Karkinos could easily access the right support channels to integrate with Google Cloud and discover the best solutions to build its proprietary tools.
These tools are used to onboard and integrate with

partner clinics to offer oncology services closer to the patient in some of the most remote areas in India.

Partners include small clinics staffed by doctors without oncological training, such as dentists, to small hospitals. The first partner of this kind was onboarded on Karkinos' platform in August 2021. Since then, over 70 partners have been signed onto the platform, who now have access to the right treatment protocol, can discover the latest treatment guidelines, and easily manage patient records closer to the patient's home. The Karkinos platform is also being built for low and middle income countries with similar challenges in mind.

Scaling to meet the demands of nationwide cancer screening

Kommu explains that Karkinos chose Google Cloud because its lower-level design needs were better supported with Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine). Its automatic scaling capabilities helped Kommu and his team optimize the platform's infrastructure during a time when computing demands were uncertain as they were in uncharted territory. It also gave developers the flexibility to experiment with all their proofs of concepts in the alpha stage.

"The ability to scale extremely fast gives me peace of mind. Another benefit we can rely on is automatic backups and disaster recovery because all of Google Cloud's services are fully managed. All we need to do are some initial configurations, and then everything just works without fail."

—Raja Sekhar Kommu, Co-founder and Chief Technology Officer, Karkinos

Karkinos extensively uses Cloud Storage (https://cloud.google.com/storage) to store objects like images of patients' scans. Compute Engine (https://cloud.google.com/compute) is leveraged to run an average of 100 instances regularly to help over 150 engineers create a unified development, quality assurance, user acceptance, and production environment. Cloud SQL (https://cloud.google.com/sql) is a key solution to handle its risk assessments at scale, with Postgre and MariaDB running on this service. Then, Google Cloud Armor (https://cloud.google.com/armor) secures its entire network through scalable and intelligent WAF and DDoS protection.

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Google Cloud also provided an avenue to control long-term costs, and Karkinos' experience was further enhanced with billing support from Persistent Systems (https://www.persistent.com/partner-ecosystem/google/), who has been instrumental in helping them to consolidate all of their billing requirements and extend cash flow with access to favorable credit terms.

Continuous improvements for a pioneering oncology platform

With all 550 members of the Karkinos team on Google Workspace (https://workspace.google.com/intl/en_us/), the company is committed to creating a seamless work experience with all of its Google Cloud applications. For example, the company relies heavily on Forms (https://workspace.google.com/intl/en_us/products/forms/) for its risk assessment surveys, which feed data to Sheets (https://workspace.google.com/intl/en_us/products/sheets/).

"When we can rely on many no-code tools, this accelerates application deployment. That's the benefit from a business perspective, but we can also integrate HR processes and team collaboration using Docs (https://workspace.google.com/intl/en_us/products/docs/) and Sheets," Kommu says.

"Google Cloud is the heart and the nerve center of Karkinos' platform. Without it, we would not have a platform. We made the right choice and are very happy to have integrated with Google Cloud because of the great support from the Google Cloud team."

—Raja Sekhar Kommu, Co-founder and Chief Technology Officer, Karkinos

As Kommu and his team look to deliver more capabilities for Karkinos' platform, they are looking forward to rolling out new applications with other Google Cloud solutions.

They have already started developing automated machine learning models to analyze radiology and histopathology images to help clinicians automatically annotate images and identify tumors. Karkinos is also developing a natural language processing model to automatically convert oncology clinical notes into structured data.

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